TEAM PROMAX's PITCH DECK

WATTWISE: SOLAR SPLITS THAT FIT

A FORECAST-LED LOCAL SOLAR ENERGY ALLOCATION ENGINE

SIME DARBY HACKATHON: TOWNSHIP OF TOMORROW

TEAM PROMAX:

01 TEAM LEAD

SHERI ANNE CHONG TING TING · LEAD, FULL-STACK
UNIVERSITY OF SOUTHAMPTON MALAYSIA (UOSM)
satc1g24@soton.ac.uk
019-807 2503

02 MEMBERS

ONG HUI MIN · UI/UX DEVELOPER, RESEARCH
UOSM · hmo1e25@soton.ac.uk

JESS LAU XI EN · UI/UX DESIGNER, DEVELOPER
UOSM · xel1n24@soton.ac.uk

LIM POH SHIU · DATA ANALYST, QA
UOSM ·psl1e25@soton.ac.uk

1) PROBLEM STATEMENT

Sime Darby Property (SDP) was the first to implement Malaysia's **Community Renewable Energy Aggregation Mechanism (CREAM)**—a program launched by PETRA, regulated by the Energy Commission, and operated on TNB's grid that enables residential rooftop leasing and local clean energy sharing—through its 6,000-home pilot in Elmina alongside GSPARX. **However, rigid allocation mechanisms create structural revenue losses for CREAM aggregators and homeowners.**

01 Each buyer's solar share must be filed with 120 days' written notice, before its real demand is known (Skrine, 2025).

02 Solar allocated beyond a buyer's use goes to the grid unpaid, yet the 9 sen/kWh access charge is still paid. Held back, it could sell via NEDA at 8 sen/kWh (Energy Commission, 2025).

03 Replayed on real meters from 407 two-buyer schemes, **a fixed split gives away a median 6.1% of the year's solar, and over 16% in one scheme in ten** (Miller et al., 2020).

2) TARGET USER(S)

Who has the problem today, and what WATTWISE changes for them

01

CREAM AGGREGATORS

- Must file every buyer's share 120 days ahead, before demand is known.
- With WATTWISE, each split is sized to the buyer's predicted use, so less solar is given away.

02

ELMINA HOMEOWNERS LEASING ROOFS

- Lease their roofs but cannot see how much of their solar is actually paid for.
- With WATTWISE, less of it is given away (6.1% to 1.3%, modelled), each roof's share of solar is shown, and up to ~20% more roofs could join (Appendix C).

03

CREAM BUYERS (BUSINESSES, SCHOOLS)

- Innovation Park today; a school, petrol stations and commercial hubs have shown interest (The Malaysian Reserve, 2025).
- With WATTWISE, each share is estimated from the buyer's own meter history, so green energy matches real use.

3) OUR SOLUTION:
WATTWISE

WATTWISE is a software layer that replaces a fixed yearly guess with an evidence-based monthly plan. **It reads each buyer's own meter history, takes the same month last year, adjusts it for the latest trend, and sizes each filing to expected demand, four months ahead.**

HOW DOES IT HELP THE TARGET USERS?

ESTIMATE

- Each buyer's use is estimated from its own meter history.

ALLOCATE

- Each share is sized to that estimate; a slice is held back and sold, so far less is given away.

LEARN

- Each month is checked against real use, and the next estimate re-bases.

RESULTS

- Splits stamped with their 120-day notice date, ready for SDP to file, and a statement for every buyer.

4) INNOVATION & DIFFERENTIATION

WHAT MAKES THIS IDEA UNIQUE, WHAT GIVES WATTWISE A NEW APPROACH?

Scope	Company	Solution
Making & storing energy	Solarvest, GSPARX, Plus Xnergy	Solar EPC, battery arrays, bundled peak-shaving software — more generation, more storage
Managing building energy use	Schneider Electric (EcoStruxure), Tanand (EasiEMS)	BMS monitoring, demand-charge warnings, HVAC optimisation — less waste inside one building
Onboarding roofs into solar programmes	Bright Energy (cream.solar), GSPARX	Sign families up, lease roofs, install panels. Their job ends when the scheme starts generating.
Dynamic CREAM energy allocation	WATTWISE	Each buyer's share must be filed 120 days ahead — WattWise plans it from the buyers' own meter history.

WATTWISE works WITH solutions within the CREAM ecosystem, not against them.

5) HOW IT WORKS

THEME: PLANET
SMART ENERGY EFFICIENCY

1. READ

- Three files are given:
  - Buyer’s monthly meter readings
  - Splits already filed
  - Scheme's solar output

2. ESTIMATE

- WATTWISE estimates each buyer’s consumption 120 days ahead.
- The estimator only ever sees data that existed before the allocation release date.

3. ALLOCATE

- Each buyer's share is set to its estimate.
- The rest is held back and sold off at 8 sen/kWh

4. FILE

- Each split is stamped with its 120-day notice date.

5. LEARN

- TNB bills the month on real meters.
- Next: suggested vs filed vs actual is checked and every buyer gets a statement.

6) IMPACT

WHAT POSITIVE IMPACT CAN WATTWISE CREATE?

01

MAXIMIZED SCHEME REVENUE

- **Reduces solar giveaway**: Cuts unallocated solar giveaway from **6.1% down to 1.3%** across 407 real two-buyer schemes. (Appendix A)
- **Direct financial gain**: Adds an estimated **RM108k – RM265k/year** to scheme income at 6,000 homes (2 buyers assumed), modelled at 20–30 sen/kWh. (Appendix B)

02

EXPANDED RESIDENT PARTICIPATION

- **20% more CREAM contributors**: Allows SDP to build solar schemes **20% larger** for the same buyers, still giving away less than a fixed split. (Appendix C)
- **Broader community impact**: Enables **more families** in Elmina to lease their roofs and earn passive green income.

03

STRATEGIC ALIGNMENT & FUTURE SCALE

- **Accelerates ESG targets**: Supports SDP's goal of a **40% Scope 1+2 emissions reduction by 2030** and Net Zero by 2050.
- **Builds a long-term data asset**: Buyers' meter and settlement data creates a foundation for allocation planning across all SDP townships.

7) FEASIBILITY

CAN OUR IDEA REALISTICALLY BE IMPLEMENTED?

01

PROVEN SOFTWARE READINESS

- **Working Prototype:** Reads the buyers' meter-readings CSV and outputs suggested allocations with 120-day notice dates.
- **Rigorously Validated:** Tested across 821 real buildings over a 4-month planning gap, giving away less than fixed splits and naive copying. (Appendix D)

02

ZERO-HARDWARE DEPLOYMENT

- **No Infrastructure Cost**: Operates purely as a software layer with **zero hardware requirements** or **upfront IT integration**.

03

STRATEGIC ALIGNMENT

- **Net-Zero Integration**: Targets SDP's largest emissions source (**Scope 2 electricity, 19,290 tCO2e**), supporting 2030 and net-zero goals. (Sime Darby Property, 2026b)
- **Low-Friction Adoption**: Matches SDP's published strategy to scale digital energy services via **risk-free validation steps**. (Sime Darby Property, 2026a)

7) IMPLEMENTATION

HOW WILL WATTWISE BE IMPLEMENTED INTO SDP'S PIPELINE?

01

NO-COST SHADOW PILOT

3 months, free. SDP shares 3 files; the first shadow report is back within a day.

02

VERIFY AGAINST REALITY

Compare suggested, filed and billed splits on measures agreed upfront.

03

LIVE AT ELMINA

WattWise suggests the real splits; SDP still files. Every buyer gets a monthly statement.

04

AUTOMATED SCALE AT ELMINA

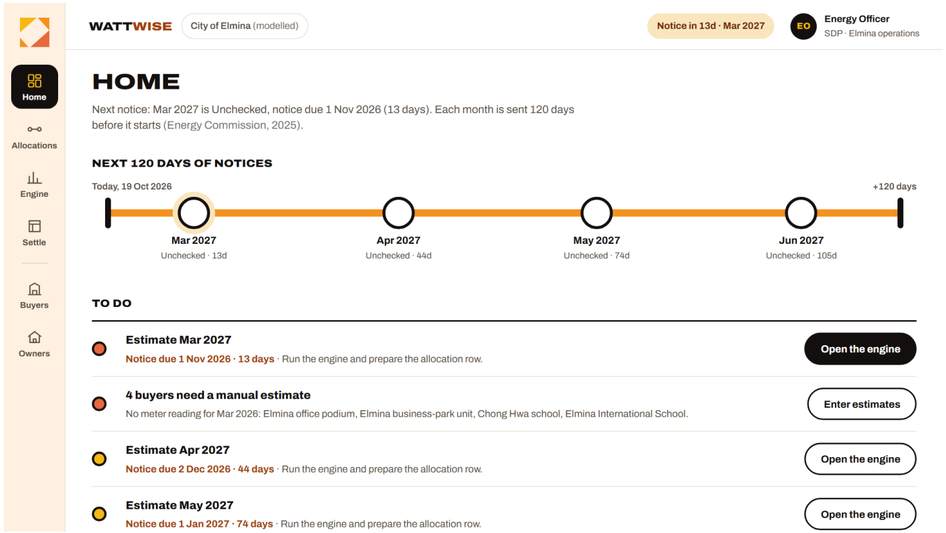
Monthly runs for each new buyer, run automatically.

05

NETWORK EXPANSION ACROSS PENINSULAR MALAYSIA

Same 3 files for every relevant township and any approved CREAM aggregator on TNB's grid.

7) VISUAL CONCEPT



OPEN THE LIVE DASHBOARD

wattwise-promax.netlify.app

TESTING SUITE FOR JUDGES

every claim, re-run on real meters

Explore the working prototype:

- GitHub: ttc-099/Wattwise
- Demo video
- Testing suite (live)

8) WHY WE SHOULD WIN

01



JESS LAU XI EN
UI/UX DESIGNER/DEVELOPER

Malaysia's new CREAM framework is an uncharted frontier for community solar, making efficient energy usage an immediate commercial necessity!

02



LIM POH SHIU
DATA ANALYST/QA

We target the complex data challenges of this unexplored niche, proving that smart predictive forecasts can help prevent clean energy wastage.

03



AMANDA ONG HUI MIN
UI/UX DEVELOPER/RESEARCH

Slashing that energy leakage makes CREAM much healthier, which means we could welcome up to 20 percent more neighborhood family roofs as contributors to a cleaner Malaysia.

04



SHERI ANNE CHONG TING TING
FULL-STACK/PROJECT LEAD

We are ready to help SDP pioneer the CREAM scheme further, creating decentralized green townships that are scalable, efficient, and rewarding for everyone involved.

FROM TEAM PROMAX,

THANK YOU FOR YOUR TIME!

Completed: 23 Oct 2026

APPENDIX A: DOES IT REALLY REDUCE SOLAR GIVEAWAY?

Result: across 407 two-buyer schemes built from 821 real buildings, a fixed split gives away a median 6.05% of the year's solar. WattWise gives away 1.34%, and less than a fixed split in 99.8% of schemes. (Miller et al., 2020; European Commission JRC, 2022; WattWise, 2026)

TEST SIMULATION

APPENDIX B: DO HOMEOWNERS REALLY EARN THAT MUCH?

Result: at 20 to 30 sen/kWh, WattWise adds RM108k to RM265k a year to a two-buyer scheme of 6,000 homes (41,265 MWh of solar), 1.4% to 6.6% of its income after the access charge, or RM25k to RM60k at Elmina's modelled size (1,368 roofs). Modelled; the CREAM price is private. (WattWise, 2026)

TEST SIMULATION

APPENDIX C: WILL THERE REALLY BE SPACE FOR 20% MORE CREAM CONTRIBUTORS?

Result: at 1.2× scheme size, WattWise gives away a median 2.17%, still below the 6.05% a fixed split gives away today. About 76% of the extra solar is sold via NEDA at 8 sen, and feeder limits and roof availability are not modelled. (WattWise, 2026)

TEST SIMULATION

APPENDIX D: DYNAMIC SPLITTING WORKS BETTER THAN FIXED ONES?

Result: median solar given away: fixed split 6.05%, copying last year 1.95%, WattWise 1.34%. WattWise gives away less than copying last year in 83.3% of two-buyer schemes; above about 20 sen, copying last year earns slightly more.

(Miller et al., 2020; WattWise, 2026)

TEST SIMULATION

APPENDIX E: REFERENCES

Energy Commission (2025) Guidelines for Community Renewable Energy Aggregation Mechanism (CREAM), GP/ST/No.55/2025. Putrajaya: Suruhanjaya Tenaga. Available at: https://www.st.gov.my (Accessed: 18 October 2026).

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WattWise (2026) WattWise code and results [Computer program]. Available at: https://github.com/ttc-099/Wattwise (Accessed: 18 October 2026).

Every figure in this deck is rebuilt by one command: python replay.py in 02 Scheduler Module of the repository.

TEST SIMULATION